HTML & CSS Ultimate Reference Guide

Complete cheatsheet with every essential concept, property, and technique

1. HTML Document Structure

Basic HTML5 Boilerplate Foundation

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="Page description for SEO">
    <title>Page Title</title>
    <link rel="stylesheet" href="styles.css">
    <link rel="icon" href="favicon.ico">
</head>
<body>
    <!-- Content goes here -->
    <script src="app.js"></script>
</body>
</html>
```

<!DOCTYPE html> tells the browser this is HTML5. Always include it as the very first line. The viewport meta tag is essential for responsive design.

Common Meta Tags <meta>

```
<!-- Character encoding -->
<meta charset="UTF-8">

<!-- Responsive viewport -->
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<!-- SEO description (150-160 chars) -->
<meta name="description" content="A concise page description">

<!-- Author -->
<meta name="author" content="John Doe">

<!-- Open Graph (social sharing) -->
<meta property="og:title" content="Page Title">
<meta property="og:description" content="Description">
<meta property="og:image" content="image.jpg">

<!-- Twitter Card -->
<meta name="twitter:card" content="summary_large_image">

<!-- Theme color for mobile browsers -->
<meta name="theme-color" content="#3b82f6">

<!-- Prevent caching -->
<meta http-equiv="Cache-Control" content="no-cache">
```

2. HTML Text Elements

Headings & Paragraphs <h1> to <p>

```
<h1>Main Page Title (use once per page)</h1>
<h2>Section Heading</h2>
<h3>Subsection Heading</h3>
<h4>Smaller Heading</h4>
<h5>Minor Heading</h5>
<h6>Smallest Heading</h6>

<p>This is a paragraph. Use <strong>bold</strong> for importance
and <em>italic</em> for emphasis.</p>

<p>Line break with <br> starts a new line.</p>

<hr>  <!-- Horizontal rule (thematic break) -->
```

RENDERED OUTPUT

Main Page Title

Section Heading

Subsection Heading

This is a paragraph with **bold** and *italic* text.

SEO Rule: Use only ONE <h1> per page. Headings should follow hierarchy (h1 → h2 → h3) without skipping levels.

Text Formatting Tags Inline Elements

Tag	Purpose	Example
	Strong importance (bold)	**Bold text**
	Emphasis (italic)	*Italic text*
<mark>	Highlighted text	Highlighted
<del>	Deleted/strikethrough	~~Deleted~~
<ins>	Inserted/underlined	Inserted
<sub>	Subscript	H_2O
<sup>	Superscript	x^2
<code>	Inline code	`print()`
<kbd>	Keyboard input	`Ctrl+C`
<samp>	Sample output	`Error: 404`
<var>	Variable in math/code	$E = mc^2$
<abbr>	Abbreviation	HTML
<cite>	Citation/title	*The Great Gatsby*
<q>	Short inline quotation	“To be or not to be”
<blockquote>	Long quotation (block)	— See below —
<pre>	Preformatted text	— Preserves whitespace —

```
<blockquote cite="https://example.com">
    <p>The only way to do great work is to love what you do.</p>
    <footer>— <cite>Steve Jobs</cite></footer>
</blockquote>

<pre>
function hello() {
    console.log("Hello, World!");
}
</pre>
```

Line Breaks & Horizontal Rules
, <hr>, <wbr>

```
<p>This text has a<br>line break here.</p>

<hr>  <!-- Thematic break between sections -->

<!-- Word break opportunity (allows break at specific point) -->
<p>supercalifragilisticexpialidocious</p>
<p>supercali<wbr>fragilistic<wbr>expialidocious</p>
```

> Use
 sparingly. For spacing, prefer CSS margin/padding. Use <hr> to indicate a thematic break in content, not just for visual lines (use CSS borders for that).

3. Links & Navigation

Anchor Tag <a> - Complete Guide <a href="">

```
<!-- External link (absolute URL) -->
<a href="https://www.example.com">Visit Example</a>

<!-- Open in new tab (use rel for security) -->
<a href="https://example.com" target="_blank" rel="noopener noreferrer">
    Open in New Tab
</a>

<!-- Internal link (relative URL) -->
<a href="about.html">About Page</a>
<a href="/contact">Contact (root-relative)</a>
<a href="../index.html">Go Up One Level</a>

<!-- Link to page section (anchor) -->
<a href="#section-2">Jump to Section 2</a>
<!-- Target: --> <section id="section-2">...</section>

<!-- Email link -->
<a href="mailto:someone@example.com">Send Email</a>
<a href="mailto:help@site.com?subject=Support Request&body=Hello,">
    Email with Subject
</a>

<!-- Phone link (mobile) -->
<a href="tel:+1234567890">Call Us</a>

<!-- SMS link -->
<a href="sms:+1234567890?body=Hello">Send SMS</a>

<!-- Download file -->
<a href="document.pdf" download>Download PDF</a>
<a href="image.jpg" download="vacation-photo.jpg">
    Download with Custom Name
</a>

<!-- Link that triggers JavaScript -->
<a href="javascript:void(0);" onclick="myFunction()">Click Me</a>
<a href="#" onclick="event.preventDefault(); doSomething();">Better Way</a>
```

> **Security:** Always use `rel="noopener noreferrer"` with `target="_blank"` to prevent tabnabbing attacks and improve performance.

Navigation with <nav> Semantic

```
<nav aria-label="Main navigation">
    <ul>
        <li><a href="/" aria-current="page">Home</a></li>
        <li><a href="/about">About</a></li>
        <li><a href="/services">Services</a></li>
        <li><a href="/contact">Contact</a></li>
    </ul>
</nav>

<!-- Breadcrumb navigation -->
<nav aria-label="Breadcrumb">
    <ol>
        <li><a href="/">Home</a></li>
        <li><a href="/products">Products</a></li>
        <li aria-current="page">Laptops</li>
    </ol>
</nav>
```

> Use `aria-label` to distinguish multiple <nav> elements. Use `aria-current="page"` to indicate the active page for screen readers.

4. Images & Media

Images - Complete Reference `<img>`

```
<!-- Basic image (always include alt!) -->
<img src="photo.jpg" alt="Descriptive text for accessibility">

<!-- With width and height (prevents layout shift / CLS) -->
<img src="photo.jpg" alt="Description" width="800" height="600">

<!-- Responsive image -->
<img src="photo.jpg" alt="Description"
     style="max-width: 100%; height: auto; display: block;">

<!-- Figure with caption -->
<figure>
    <img src="chart.png" alt="Sales chart showing Q1 growth">
    <figcaption>Figure 1: Q1 Sales Growth (2026)</figcaption>
</figure>

<!-- Lazy loading (loads when scrolled into view) -->
<img src="below-fold.jpg" alt="Description" loading="lazy">

<!-- Eager loading (loads immediately - for above-fold images) -->
<img src="hero.jpg" alt="Hero image" loading="eager">

<!-- Responsive images with srcset -->
<img src="image-800.jpg"
     srcset="image-400.jpg 400w,
             image-800.jpg 800w,
             image-1200.jpg 1200w"
     sizes="(max-width: 600px) 400px,
            (max-width: 1000px) 800px,
            1200px"
     alt="Responsive image">

<!-- Picture element for art direction -->
<picture>
    <source srcset="mobile.jpg" media="(max-width: 600px)">
    <source srcset="tablet.jpg" media="(max-width: 1000px)">
    <img src="desktop.jpg" alt="Responsive image">
</picture>
```

> **Accessibility:** The `alt` attribute is required. Use empty alt (`alt=""`) for decorative images. Never omit it entirely.

Audio & Video `<audio>, <video>`

```
<!-- Audio with multiple formats -->
<audio controls preload="metadata">
    <source src="music.mp3" type="audio/mpeg">
    <source src="music.ogg" type="audio/ogg">
    <p>Your browser does not support audio.
       <a href="music.mp3">Download the file</a>.</p>
</audio>

<!-- Video with poster, captions, and multiple sources -->
<video width="640" height="360" controls
       poster="thumbnail.jpg" preload="none">
    <source src="movie.mp4" type="video/mp4">
    <source src="movie.webm" type="video/webm">

    <track kind="subtitles" src="subtitles-en.vtt"
           srclang="en" label="English" default>
    <track kind="subtitles" src="subtitles-es.vtt"
           srclang="es" label="Spanish">

    <p>Your browser does not support video.</p>
</video>

<!-- Attributes: controls, autoplay, loop, muted, playsinline -->
```

Attribute	Description
controls	Shows play/pause/volume controls
autoplay	Starts playing automatically (muted required in most browsers)
loop	Loops playback
muted	Mutes audio by default
preload	none | metadata | auto
poster	Image shown before video plays
playsinline	Prevents fullscreen on iOS

Embedding Content (iframe, embed, object) `<iframe>`

```
<!-- Embed another webpage -->
<iframe src="https://example.com"
        width="600" height="400"
        title="Description of embedded content"
        loading="lazy">
</iframe>

<!-- YouTube embed -->
<iframe src="https://www.youtube.com/embed/VIDEO_ID"
        width="560" height="315"
        title="YouTube video"
        frameborder="0"
        allow="accelerometer; autoplay; clipboard-write;
               encrypted-media; gyroscope; picture-in-picture"
        allowfullscreen>
</iframe>

<!-- Google Maps -->
<iframe src="https://maps.google.com/maps?q=LOCATION&output=embed"
        width="600" height="450" style="border:0;">
</iframe>

<!-- Secure iframe with sandbox -->
<iframe src="untrusted-content.html"
        sandbox="allow-scripts allow-same-origin">
</iframe>
```

> Always include a `title` attribute on iframes for accessibility. Use the `sandbox` attribute to restrict capabilities of embedded content for security.

5. Lists

All List Types <ul>, <ol>, <dl>

```
<!-- Unordered List (bullets) -->
<ul>
    <li>Apples</li>
    <li>Bananas</li>
    <li>Oranges</li>
</ul>

<!-- Ordered List (numbers) -->
<ol>
    <li>First step</li>
    <li>Second step</li>
    <li>Third step</li>
</ol>

<!-- Ordered with type and start -->
<ol type="A" start="3">  <!-- C, D, E -->
    <li>Item C</li>
    <li value="10">Item 10 (custom number)</li>
    <li>Item 11</li>
</ol>

<!-- Reversed ordered list -->
<ol reversed>
    <li>Step 3</li>
    <li>Step 2</li>
    <li>Step 1</li>
</ol>

<!-- Nested Lists -->
<ul>
    <li>Fruits
        <ul>
            <li>Apples</li>
            <li>Bananas</li>
        </ul>
    </li>
    <li>Vegetables
        <ul>
            <li>Carrots</li>
            <li>Broccoli</li>
        </ul>
    </li>
</ul>

<!-- Description List (name-value pairs) -->
<dl>
    <dt>HTML</dt>
    <dd>HyperText Markup Language</dd>

    <dt>CSS</dt>
    <dd>Cascading Style Sheets</dd>

    <dt>JavaScript</dt>
    <dd>Programming language for the web</dd>
</dl>
```

Attribute	Values	Description
type	1, A, a, I, i	Numbering style for
start	number	Starting number
reversed	—	Reverse numbering
value	number	Custom number for

6. Tables

Complete Table Structure `<table>`

```
<table>
    <caption>Monthly Sales Report</caption>

    <colgroup>
        <col style="width: 30%">
        <col style="width: 35%">
        <col style="width: 35%">
    </colgroup>

    <thead>
        <tr>
            <th scope="col">Month</th>
            <th scope="col">Revenue</th>
            <th scope="col">Expenses</th>
        </tr>
    </thead>

    <tbody>
        <tr>
            <th scope="row">January</th>
            <td>$10,000</td>
            <td>$6,000</td>
        </tr>
        <tr>
            <th scope="row">February</th>
            <td>$12,000</td>
            <td>$7,000</td>
        </tr>
    </tbody>

    <tfoot>
        <tr>
            <th scope="row">Total</th>
            <td>$22,000</td>
            <td>$13,000</td>
        </tr>
    </tfoot>
</table>

<!-- Merging cells -->
<table>
    <tr>
        <th colspan="2">Header spans 2 columns</th>
    </tr>
    <tr>
        <td rowspan="2">Spans 2 rows</td>
        <td>Cell 2</td>
    </tr>
    <tr>
        <td>Cell 3</td>
    </tr>
</table>
```

> **Accessibility:** Always use `scope="col"` or `scope="row"` on <th> elements so screen readers understand the table structure. Use <caption> to describe the table.

7. Forms & Input Elements

Complete Form Structure `<form>`

```
<form action="/submit" method="POST"
      enctype="multipart/form-data" novalidate>

    <!-- Text input with label -->
    <label for="username">Username:</label>
    <input type="text" id="username" name="username"
           placeholder="Enter username" required
           minlength="3" maxlength="20"
           pattern="[A-Za-z0-9_]+">

    <!-- Email -->
    <label for="email">Email:</label>
```

```
    <input type="email" id="email" name="email"
           placeholder="you@example.com" required>

    <!-- Password -->
    <label for="password">Password:</label>
    <input type="password" id="password" name="password"
           minlength="8" required>

    <!-- Number with range -->
    <label for="age">Age:</label>
    <input type="number" id="age" name="age"
           min="18" max="120" step="1" value="25">

    <!-- Telephone -->
    <label for="phone">Phone:</label>
    <input type="tel" id="phone" name="phone"
           pattern="[0-9]{3}-[0-9]{3}-[0-9]{4}"
           placeholder="123-456-7890">

    <!-- URL -->
    <label for="website">Website:</label>
    <input type="url" id="website" name="website"
           placeholder="https://example.com">

    <!-- Search -->
    <label for="search">Search:</label>
    <input type="search" id="search" name="search"
           placeholder="Search...">

    <!-- Date and Time inputs -->
    <label for="birthday">Birthday:</label>
    <input type="date" id="birthday" name="birthday">

    <label for="meeting">Meeting Time:</label>
    <input type="datetime-local" id="meeting" name="meeting">

    <label for="month">Month:</label>
    <input type="month" id="month" name="month">

    <label for="week">Week:</label>
    <input type="week" id="week" name="week">

    <label for="appt">Time:</label>
    <input type="time" id="appt" name="appt">

    <!-- Color picker -->
    <label for="favcolor">Favorite Color:</label>
    <input type="color" id="favcolor" name="favcolor" value="#3b82f6">

    <!-- Range slider -->
    <label for="volume">Volume: <span id="vol-value">50</span>%</label>
    <input type="range" id="volume" name="volume"
           min="0" max="100" value="50">

    <!-- File upload (requires enctype="multipart/form-data") -->
    <label for="resume">Upload Resume:</label>
    <input type="file" id="resume" name="resume"
           accept=".pdf,.doc,.docx" multiple>

    <!-- Hidden field -->
    <input type="hidden" name="form-id" value="contact-123">

    <!-- Textarea -->
    <label for="message">Message:</label>
    <textarea id="message" name="message" rows="5" cols="50"
              placeholder="Enter your message..."></textarea>

    <!-- Select dropdown -->
    <label for="country">Country:</label>
    <select id="country" name="country" required>
        <option value="" disabled selected>Select a country</option>
        <option value="us">United States</option>
        <option value="uk">United Kingdom</option>
        <option value="ca">Canada</option>
    </select>

    <!-- Select with option groups -->
    <label for="car">Car:</label>
    <select id="car" name="car">
        <optgroup label="German Cars">
            <option value="mercedes">Mercedes</option>
            <option value="audi">Audi</option>
        </optgroup>
        <optgroup label="Japanese Cars">
            <option value="toyota">Toyota</option>
```

```
            <option value="honda">Honda</option>
        </optgroup>
    </select>

    <!-- Datalist (autocomplete suggestions) -->
    <label for="browser">Browser:</label>
    <input list="browsers" id="browser" name="browser">
    <datalist id="browsers">
        <option value="Chrome">
        <option value="Firefox">
        <option value="Safari">
        <option value="Edge">
    </datalist>

    <!-- Checkboxes -->
    <fieldset>
        <legend>Interests:</legend>
        <label><input type="checkbox" name="interests" value="sports"> Sports</label>
        <label><input type="checkbox" name="interests" value="music" checked> Music</label>
        <label><input type="checkbox" name="interests" value="reading"> Reading</label>
    </fieldset>

    <!-- Radio buttons -->
    <fieldset>
        <legend>Gender:</legend>
        <label><input type="radio" name="gender" value="male" required> Male</label>
        <label><input type="radio" name="gender" value="female"> Female</label>
        <label><input type="radio" name="gender" value="other"> Other</label>
    </fieldset>

    <!-- Progress and Meter -->
    <label>Upload Progress:</label>
    <progress value="70" max="100">70%</progress>

    <label>Disk Usage:</label>
    <meter value="0.7" min="0" max="1" low="0.5" high="0.8" optimum="0.3">70%</meter>

    <!-- Output (for calculation results) -->
    <label>Result: <output name="result" for="a b">0</output></label>

    <!-- Buttons -->
    <button type="submit">Submit</button>
    <button type="reset">Reset</button>
    <button type="button" onclick="doSomething()">Click Me</button>

</form>
```

> **Best Practices:** Always associate <label> with inputs using `for` attribute matching the input's `id`. Use `fieldset` + `legend` to group related controls. The `novalidate` attribute disables browser validation (useful when using custom JS validation).

Form Validation Attributes HTML5 Validation

Attribute	Description	Example
required	Field must be filled	<input required>
minlength	Minimum character length	<input minlength="3">
maxlength	Maximum character length	<input maxlength="50">
min	Minimum numeric/date value	<input min="0">
max	Maximum numeric/date value	<input max="100">
step	Increment step	<input step="5">
pattern	Regex pattern	<input pattern="[A-Z]{3}">
readonly	Non-editable but submitted	<input readonly>
disabled	Non-editable, not submitted	<input disabled>
placeholder	Hint text	<input placeholder="Name">
autofocus	Focus on page load	<input autofocus>
autocomplete	Browser autofill	<input autocomplete="email">
multiple	Allow multiple values	<input multiple>
accept	Allowed file types	<input accept=".jpg,.png">
novalidate	Disable validation (on form)	<form novalidate>
form	Associate with form by ID	<input form="myform">
formaction	Override form action	<button formaction="/save">

8. Semantic HTML5 Elements

Page Structure Elements Layout

```
<!-- Complete semantic page structure -->
<body>
    <header>
        <nav aria-label="Main">
            <ul>
                <li><a href="/">Home</a></li>
                <li><a href="/about">About</a></li>
            </ul>
        </nav>
    </header>

    <main>
        <article>
            <header>
                <h1>Article Title</h1>
                <time datetime="2026-01-15">January 15, 2026</time>
            </header>

            <section>
                <h2>Introduction</h2>
                <p>Content here...</p>
            </section>

            <section>
                <h2>Main Content</h2>
                <p>More content...</p>
                <figure>
                    <img src="chart.jpg" alt="Data chart">
```

```
                    <figcaption>Figure 1: Annual data</figcaption>
                </figure>
            </section>

            <footer>
                <p>Written by <address><a href="mailto:author@site.com">Author Name</a></address></p>
            </footer>
        </article>

        <aside>
            <h3>Related Articles</h3>
            <ul>
                <li><a href="/article-2">Related Post 1</a></li>
                <li><a href="/article-3">Related Post 2</a></li>
            </ul>
        </aside>
    </main>

    <footer>
        <p>&copy; 2026 Company Name. All rights reserved.</p>
        <nav aria-label="Footer">
            <a href="/privacy">Privacy Policy</a>
            <a href="/terms">Terms of Service</a>
        </nav>
    </footer>
</body>
```

Element	Purpose	Usage
<header>	Introductory content	Page header, article header, section header
<nav>	Navigation links	Main menu, footer links, breadcrumbs
<main>	Main content (unique per page)	Only one per page
<article>	Self-contained content	Blog posts, news articles, comments
<section>	Thematic grouping	Chapters, tabbed content, distinct sections
<aside>	Sidebar content	Related links, ads, pull quotes
<footer>	Footer content	Copyright, author info, related links
<address>	Contact information	Author/owner contact details
<time>	Date/time	<time datetime="2026-01-15">
<mark>	Highlighted text	Search results, important text
<details>	Disclosure widget	FAQ, collapsible content
<summary>	Details summary	Always first child of <details>
<dialog>	Modal dialog box	Popups, modals (requires JS)

Details, Summary & Dialog Interactive

```
<!-- Accordion / FAQ (no JavaScript needed) -->
<details>
    <summary>What is HTML?</summary>
    <p>HTML (HyperText Markup Language) is the standard markup language for documents designed to be displayed
in a web browser.</p>
</details>

<details open>  <!-- Open by default -->
    <summary>What is CSS?</summary>
    <p>CSS (Cascading Style Sheets) is used to style and layout web pages.</p>
</details>

<!-- Dialog (modal) - requires JavaScript -->
<dialog id="myDialog">
    <h2>Confirm Action</h2>
    <p>Are you sure you want to proceed?</p>
    <button onclick="document.getElementById('myDialog').close()">
        Cancel
    </button>
    <button onclick="document.getElementById('myDialog').close()">
        Confirm
    </button>
</dialog>

<button onclick="document.getElementById('myDialog').showModal()">
    Open Dialog
</button>
```

<details> and <summary> create native accordions without JavaScript. <dialog> creates accessible modal dialogs with built-in focus management and backdrop.

9. Global HTML Attributes

Attributes Available on ALL HTML Elements Universal

Attribute	Description	Example
accesskey	Keyboard shortcut	Home (Alt+H)
class	CSS class name(s)	
contenteditable	Editable content	
data-*	Custom data attributes	
dir	Text direction	Right to left
draggable	Draggable element	<img draggable="true">
hidden	Hides element	<div hidden>Hidden
id	Unique identifier	<section id="about">
lang	Language code	Hola
spellcheck	Spell checking	<textarea spellcheck="true">
style	Inline CSS	
tabindex	Tab order	(0=focusable, -1=programmatic)
title	Tooltip text	<abbr title="HyperText">HT</abbr>
translate	Translation behavior	<code translate="no">console.log()</code>

```
<!-- Data attributes (custom data) -->
<button class="btn"
        data-action="delete"
        data-id="42"
        data-confirm="Are you sure?">
    Delete Item
</button>

<!-- Access in JavaScript -->
<script>
    const btn = document.querySelector('.btn');
    console.log(btn.dataset.action);  // "delete"
    console.log(btn.dataset.id);      // "42"
    console.log(btn.dataset.confirm); // "Are you sure?"
</script>
```

ARIA Accessibility Attributes a11y

```
<!-- Landmark roles -->
<header role="banner">...</header>
<nav role="navigation">...</nav>
<main role="main">...</main>
<aside role="complementary">...</aside>
<footer role="contentinfo">...</footer>

<!-- Widget roles -->
<div role="button" tabindex="0" aria-pressed="false">
    Toggle
</div>

<!-- Live regions (announce changes) -->
<div role="status" aria-live="polite">
    <!-- Screen readers announce updates here -->
</div>

<!-- Labels and descriptions -->
<input type="text" aria-label="Search" aria-describedby="search-hint">
<span id="search-hint">Enter keywords to search</span>

<!-- Expanded/collapsed states -->
<button aria-expanded="false" aria-controls="menu">
    Menu
</button>
<ul id="menu" hidden>...</ul>

<!-- Current page indication -->
<nav>
    <a href="/" aria-current="page">Home</a>
    <a href="/about">About</a>
</nav>

<!-- Hidden from screen readers -->
<img src="decorative.jpg" alt="" aria-hidden="true">
```

Attribute	Purpose
aria-label	Defines a label for the element
aria-labelledby	References another element as label
aria-describedby	References descriptive text
aria-hidden	Hides from assistive tech (true/false)
aria-expanded	Indicates expanded state (true/false)
aria-live	Announces updates (off/polite/assertive)
aria-current	Indicates current item (page/step/date)
aria-disabled	Indicates disabled state
aria-required	Indicates required field
role	Defines element's purpose/behavior

10. CSS Selectors

Basic Selectors Foundation

Selector	Example	Selects
Element	p	All <p> elements
Class	.intro	All elements with class="intro"
ID	#header	The element with id="header"
Universal	*	All elements
Grouping	h1, h2, p	All h1, h2, and p elements

Combinator Selectors Relationships

Selector	Example	Selects
Descendant	div p	All <p> inside (at any depth)
Child	ul > li	Direct children of
Adjacent Sibling	h2 + p	<p> immediately after <h2>
General Sibling	h2 ~ p	All <p> siblings after <h2>

```
<!-- HTML Structure -->
<article>
    <h2>Title</h2>
    <p>First paragraph (adjacent to h2)</p>
    <p>Second paragraph (general sibling)</p>
    <div>
        <p>Nested paragraph (descendant)</p>
    </div>
</article>

<!-- CSS -->
article p { color: blue; }        /* All paragraphs in article */
article > p { font-size: 18px; } /* Only direct children */
h2 + p { margin-top: 0; }        /* First paragraph after h2 */
h2 ~ p { color: gray; }          /* All paragraphs after h2 */
```

Attribute Selectors [attr]

Selector	Example	Selects
[attr]	[target]	Elements with target attribute
[attr=value]	[type="text"]	type exactly equals "text"
[attr~=value]	[class~="btn"]	class contains "btn" word
[attr|=value]	[lang|="en"]	lang is "en" or starts with "en-"
[attr^=value]	[href^="https"]	href starts with "https"
[attr$=value]	[href$=".pdf"]	href ends with ".pdf"
[attr*=value]	[href*="google"]	href contains "google"

```
/* Style all external links */
a[href^="http"] { color: green; }

/* Style all download links */
a[href$=".pdf"], a[href$=".zip"] {
    background: url(download-icon.png) no-repeat right;
    padding-right: 20px;
}

/* Style inputs by type */
input[type="email"] { border-color: blue; }
input[type="password"] { border-color: red; }

/* Case-insensitive match */
a[href*="example" i] { color: purple; }
```

Pseudo-Classes (State & Position) :hover, :nth-child()

```
/* Link states (LVHA order matters!) */
a:link { color: blue; }      /* Unvisited link */
a:visited { color: purple; } /* Visited link */
a:hover { color: red; }      /* Mouse over */
a:active { color: green; }   /* Clicking down */

/* Focus states (accessibility) */
input:focus { outline: 2px solid #3b82f6; }
button:focus-visible { outline: 3px solid orange; }

/* Form states */
input:enabled { background: white; }
input:disabled { background: #e2e8f0; }
input:checked + label { font-weight: bold; }
input:valid { border-color: green; }
input:invalid { border-color: red; }
input:required { border-left: 3px solid red; }
input:optional { border-left: 3px solid gray; }
input:placeholder-shown { color: #94a3b8; }

/* Structural pseudo-classes */
li:first-child { font-weight: bold; }
li:last-child { border-bottom: none; }
li:nth-child(3) { color: blue; }           /* 3rd child */
li:nth-child(odd) { background: #f8fafc; } /* 1st, 3rd, 5th... */
li:nth-child(even) { background: white; }   /* 2nd, 4th, 6th... */
li:nth-child(3n+1) { color: red; }         /* 1st, 4th, 7th... */
li:nth-last-child(2) { color: orange; }    /* 2nd from last */

/* Only child */
p:only-child { font-size: 20px; }

/* Empty elements */
div:empty { display: none; }

/* Not selector */
button:not(.primary) { background: gray; }
input:not([type="submit"]) { border: 1px solid #ccc; }

/* Is selector (modern browsers) */
:is(h1, h2, h3) a { color: inherit; }

/* Where selector (zero specificity) */
:where(.card, .modal) h2 { margin-top: 0; }

/* Has selector (parent selector - modern) */
article:has(img) { padding-top: 0; }
figure:has(figcaption) { border: 1px solid #ccc; }
```

> **LVHA Order:** Always define link states in this order: :link → :visited → :hover → :active. Otherwise styles may override incorrectly.

Pseudo-Elements (::before, ::after, etc.) ::

Pseudo-Element	Description
::before	Inserts content before element
::after	Inserts content after element
::first-line	First line of text
::first-letter	First letter of text
::selection	Selected/highlighted text
::placeholder	Placeholder text in inputs
::marker	List item bullets/numbers
::backdrop	Behind modal dialogs
::file-selector-button	File input button

```css
/* Decorative elements with ::before and ::after */
.quote::before {
    content: '"';
    font-size: 3em;
    color: #cbd5e1;
    position: absolute;
    left: -10px;
    top: -10px;
}

/* Required field indicator */
label.required::after {
    content: ' *';
    color: red;
}

/* Custom list bullets */
li::marker {
    color: #3b82f6;
    font-size: 1.2em;
}

/* Styled first letter (drop cap) */
p::first-letter {
    font-size: 3em;
    float: left;
    line-height: 1;
    margin-right: 8px;
    color: #1e40af;
}

/* Selection color */
::selection {
    background: #3b82f6;
    color: white;
}

/* Placeholder styling */
input::placeholder {
    color: #94a3b8;
    font-style: italic;
}

/* File upload button */
input::file-selector-button {
    background: #3b82f6;
    color: white;
    border: none;
    padding: 8px 16px;
    border-radius: 4px;
    cursor: pointer;
}
```

> ::before and ::after require the `content` property (can be empty string `""`). They create inline elements by default.

11. CSS Box Model

The Box Model Explained Core Concept

```
/* The Box Model: Content → Padding → Border → Margin */

.box {
    /* Content area */
    width: 300px;
    height: 200px;

    /* Padding (space INSIDE the border) */
    padding: 20px;                    /* All sides */
    padding: 10px 20px;               /* Vertical | Horizontal */
    padding: 10px 20px 15px;          /* Top | Horizontal | Bottom */
    padding: 10px 20px 15px 25px;     /* Top | Right | Bottom | Left (clockwise) */
    padding-top: 10px;
    padding-right: 20px;
    padding-bottom: 15px;
    padding-left: 25px;

    /* Border */
    border: 2px solid #3b82f6;       /* width style color */
    border-width: 2px;
    border-style: solid;             /* none | hidden | dotted | dashed | solid | double | groove | ridge |
inset | outset */
    border-color: #3b82f6;

    /* Individual borders */
    border-top: 3px dashed red;
    border-right: 2px solid blue;
    border-bottom: 1px dotted green;
    border-left: 4px double purple;

    /* Border radius */
    border-radius: 8px;              /* All corners */
    border-radius: 8px 16px;         /* Top-left+Bottom-right | Top-right+Bottom-left */
    border-radius: 8px 16px 24px 32px; /* TL | TR | BR | BL */
    border-radius: 50%;              /* Circle (if square) */
    border-radius: 9999px;           /* Pill shape */

    /* Margin (space OUTSIDE the border) */
    margin: 20px;                    /* All sides */
    margin: 10px auto;               /* Vertical | Horizontal (centered) */
    margin-top: 10px;
    margin-right: auto;
    margin-bottom: 10px;
    margin-left: auto;

    /* CRITICAL: Box sizing fix */
    box-sizing: border-box;          /* Width/Height includes padding + border */
    /* Without this, a 300px width + 20px padding + 2px border = 344px total! */
}

/* Global box-sizing reset (put in every project) */
*, *::before, *::after {
    box-sizing: border-box;
}
```

Always use box-sizing: border-box! Without it, padding and border are ADDED to the specified width/height, making layouts unpredictable. The global reset above is industry standard.

Outline vs Border outline

Property	Takes up space?	Affects layout?	Can be rounded?	Multiple sides?
border	Yes	Yes	Yes (border-radius)	Yes
outline	No	No	No (outline-radius not widely supported)	No (always all sides)

```
/* Outline is great for focus indicators */
input:focus {
    outline: 2px solid #3b82f6;
    outline-offset: 2px;  /* Gap between element and outline */
}

/* Remove default outline (but replace it!) */
button:focus {
    outline: none;
    box-shadow: 0 0 0 3px rgba(59, 130, 246, 0.5);
}

/* Outline shorthand */
outline: 2px dashed red;
outline-width: 2px;
outline-style: dashed;
outline-color: red;
outline-offset: 4px;
```

Width, Height & Sizing Sizing

```
/* Fixed sizing */
width: 300px;
height: 200px;

/* Percentage (of parent) */
width: 100%;
height: 50%;

/* Min/Max constraints */
min-width: 200px;      /* Cannot be smaller than 200px */
max-width: 800px;      /* Cannot be larger than 800px */
min-height: 100px;
max-height: 600px;

/* Modern responsive sizing */
width: min(100%, 1200px);          /* Smaller of 100% or 1200px */
width: max(300px, 50%);            /* Larger of 300px or 50% */
width: clamp(300px, 50%, 800px);   /* Min: 300px, Preferred: 50%, Max: 800px */

/* Fit content */
width: fit-content;     /* Shrinks to content width */
width: max-content;     /* Maximum content width (no wrapping) */
width: min-content;     /* Minimum content width (most narrow) */

/* Aspect ratio (modern) */
aspect-ratio: 16 / 9;   /* Maintains 16:9 ratio */
aspect-ratio: 1 / 1;    /* Square */

/* Object fit (for images/videos) */
img {
    width: 100%;
    height: 300px;
    object-fit: cover;       /* Fill box, crop excess */
    object-fit: contain;     /* Fit inside, letterbox */
    object-fit: fill;        /* Stretch to fit (distorts) */
    object-fit: none;        /* Original size, may clip */
    object-fit: scale-down;  /* Like contain or none, whichever is smaller */
    object-position: center; /* Position within box */
}
```

Display Property Deep Dive display

Value	Behavior	Width/Height	Margin/Padding
block	Starts new line, fills width	Respected	Full control
inline	Flows with text	Ignored	Horizontal only
inline-block	Inline flow, block properties	Respected	Full control
none	Hidden, removed from layout	—	—
flex	Flexbox container	—	—
inline-flex	Inline flexbox container	—	—
grid	Grid container	—	—
inline-grid	Inline grid container	—	—
table	Behaves like <table>	—	—
contents	Element removed, children promoted	—	—

```
/* Hide element but keep in DOM (for animations) */
.hidden { visibility: hidden; }  /* Invisible but takes up space */
.none { display: none; }          /* Removed from layout entirely */

/* Opacity vs Visibility vs Display */
.opacity-0 { opacity: 0; }       /* Invisible, still interactive (unless pointer-events: none) */
.visibility-hidden { visibility: hidden; } /* Invisible, not interactive, takes space */
.display-none { display: none; } /* Gone completely */

/* Modern: Hide but accessible to screen readers */
.visually-hidden {
    position: absolute;
    width: 1px;
    height: 1px;
    padding: 0;
    margin: -1px;
    overflow: hidden;
    clip: rect(0, 0, 0, 0);
    white-space: nowrap;
    border: 0;
}
```

12. Colors & Backgrounds

Color Formats color

```
/* Named colors (140+ available) */
color: red;
color: cornflowerblue;
color: transparent;  /* Fully transparent */

/* Hexadecimal */
color: #ff0000;      /* Red */
color: #f00;         /* Short form (duplicated digits) */
color: #ff000080;    /* With alpha (80 = 50% opacity) - 8-digit hex */
color: #f008;       /* Short with alpha */

/* RGB / RGBA */
color: rgb(255, 0, 0);
color: rgb(255 0 0);           /* Modern space-separated syntax */
color: rgba(255, 0, 0, 0.5);  /* 50% opacity (legacy) */
color: rgb(255 0 0 / 50%);    /* Modern syntax with alpha */

/* HSL / HSLA (Hue, Saturation, Lightness) */
color: hsl(0, 100%, 50%);     /* Red */
color: hsl(120 100% 50%);     /* Green (modern syntax) */
color: hsla(0, 100%, 50%, 0.5); /* With alpha (legacy) */
color: hsl(0 100% 50% / 50%);   /* Modern alpha syntax */

/* HWB (Hue, Whiteness, Blackness) - modern */
color: hwb(0 0% 0%);           /* Red */
color: hwb(0 0% 0% / 50%);    /* With alpha */

/* LAB (Lightness, A-axis, B-axis) - perceptually uniform */
color: lab(50% 50 50);

/* LCH (Lightness, Chroma, Hue) */
color: lch(50% 50 0);

/* OKLCH (improved LCH) - recommended for design systems */
color: oklch(60% 0.2 250);

/* color() function for wide-gamut displays */
color: color(display-p3 1 0 0);

/* currentColor keyword */
border: 2px solid currentColor;  /* Uses the element's text color */
```

HSL is often preferred over RGB because it's more intuitive: Hue (0-360 color wheel), Saturation (0-100% gray to full color), Lightness (0-100% black to white). **OKLCH** is the modern standard for design systems because it provides perceptually uniform lightness.

Background Properties background

```
/* Background color */
background-color: #f0f0f0;
background-color: rgba(0, 0, 0, 0.5);

/* Background image */
background-image: url('bg.jpg');
background-image: url('bg.jpg'), url('overlay.png'); /* Multiple layers */

/* Linear gradients */
background: linear-gradient(to right, #3b82f6, #8b5cf6);
background: linear-gradient(45deg, red, orange, yellow, green, blue);
background: linear-gradient(to bottom, #3b82f6 0%, #8b5cf6 50%, #ec4899 100%);
background: linear-gradient(90deg, #3b82f6 0%, #3b82f6 50%, #8b5cf6 50%, #8b5cf6 100%); /* Hard stop */

/* Radial gradients */
background: radial-gradient(circle, #3b82f6, #1e293b);
background: radial-gradient(ellipse at top, #3b82f6, transparent);
background: radial-gradient(circle at 30% 70%, #3b82f6 0%, #1e293b 100%);

/* Conic gradients */
background: conic-gradient(from 0deg, red, yellow, green, blue, red);
background: conic-gradient(from 45deg at 50% 50%, red 0deg, red 90deg, blue 90deg, blue 180deg, green 180deg,
green 270deg, yellow 270deg);

/* Repeating gradients */
background: repeating-linear-gradient(45deg, #3b82f6, #3b82f6 10px, #1e293b 10px, #1e293b 20px);
background: repeating-radial-gradient(circle, #3b82f6 0, #3b82f6 10px, #1e293b 10px, #1e293b 20px);

/* Background size */
background-size: cover;         /* Fill entire box, crop excess */
background-size: contain;       /* Fit inside box, may letterbox */
background-size: 100% 100%;     /* Stretch to fill */
background-size: 500px 300px;   /* Fixed size */
background-size: auto 100%;     /* Auto width, full height */

/* Background position */
background-position: center;              /* Center both axes */
background-position: top right;         /* Keywords */
background-position: 50% 50%;           /* Percentages */
background-position: 20px 40px;         /* Pixel values */
background-position: right 20px bottom 10px; /* Offset from edges */

/* Background repeat */
background-repeat: no-repeat;
background-repeat: repeat-x;    /* Horizontal only */
background-repeat: repeat-y;    /* Vertical only */
background-repeat: space;       /* Repeat with spacing */
background-repeat: round;       /* Repeat and stretch to fit */

/* Background attachment */
background-attachment: scroll;  /* Default - scrolls with page */
background-attachment: fixed;   /* Fixed to viewport (parallax effect) */
background-attachment: local;   /* Scrolls with element content */

/* Background origin (where background is drawn) */
background-origin: padding-box; /* Default - starts at padding edge */
background-origin: border-box;  /* Starts at border edge */
background-origin: content-box; /* Starts at content edge */

/* Background clip (where background is visible) */
background-clip: border-box;    /* Default - extends under border */
background-clip: padding-box;   /* Stops at padding edge */
background-clip: content-box;   /* Only in content area */
background-clip: text;          /* Only behind text (requires -webkit- prefix) */

/* Shorthand: color image position/size repeat origin clip attachment */
background: #f0f0f0 url('bg.jpg') center/cover no-repeat fixed;

/* Multiple backgrounds (layered) */
background:
    linear-gradient(rgba(0,0,0,0.5), rgba(0,0,0,0.5)),
    url('bg.jpg') center/cover no-repeat;

/* Text gradient (webkit only) */
background: linear-gradient(90deg, #3b82f6, #ec4899);
-webkit-background-clip: text;
-webkit-text-fill-color: transparent;
background-clip: text;
```

13. Typography

Font Properties `font-*`

```
/* Font family (always provide fallbacks) */
font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
font-family: Georgia, 'Times New Roman', Times, serif;
font-family: 'Courier New', Courier, monospace;
font-family: system-ui, -apple-system, BlinkMacSystemFont, sans-serif;

/* Font size */
font-size: 16px;
font-size: 1.5rem;        /* Relative to root font size */
font-size: 1.2em;         /* Relative to parent font size */
font-size: 2vw;           /* Relative to viewport width */
font-size: clamp(1rem, 2.5vw, 2rem);  /* Responsive font size */

/* Font weight */
font-weight: normal;      /* 400 */
font-weight: bold;        /* 700 */
font-weight: 300;         /* Light */
font-weight: 400;         /* Normal */
font-weight: 700;         /* Bold */
font-weight: 900;         /* Black/Heavy */

/* Font style */
font-style: normal;
font-style: italic;
font-style: oblique 15deg; /* Slanted at 15 degrees */

/* Font variant */
font-variant: small-caps;

/* Line height (leading) */
line-height: 1.5;         /* Unitless: multiplies by font-size (RECOMMENDED) */
line-height: 24px;        /* Fixed height */
line-height: 150%;        /* Percentage of font-size */

/* Font shorthand: style variant weight size/line-height family */
font: italic small-caps bold 16px/1.5 'Segoe UI', sans-serif;

/* Letter and word spacing */
letter-spacing: 0.05em;   /* Space between letters */
letter-spacing: -0.02em;  /* Tighter tracking */
word-spacing: 0.25em;     /* Space between words */

/* Text alignment */
text-align: left;
text-align: right;
text-align: center;
text-align: justify;       /* Even margins (be careful with readability) */
text-align-last: center;  /* Alignment of last line */

/* Text decoration */
text-decoration: underline;
text-decoration: overline;
text-decoration: line-through;
text-decoration: none;     /* Remove underline from links */

/* Modern text decoration (more control) */
text-decoration-line: underline;
text-decoration-style: wavy;     /* solid | double | dotted | dashed | wavy */
text-decoration-color: red;
text-decoration-thickness: 2px;
text-underline-offset: 4px;      /* Distance from text */

/* Text transform */
text-transform: uppercase;
text-transform: lowercase;
text-transform: capitalize;   /* First letter of each word */
text-transform: none;

/* Text indent */
text-indent: 2em;         /* Indent first line */

/* White space handling */
white-space: normal;      /* Collapses whitespace, wraps */
white-space: nowrap;      /* No wrapping */
white-space: pre;         /* Preserves whitespace (like <pre>) */
white-space: pre-wrap;    /* Preserves whitespace, wraps at line breaks */
```

```
white-space: pre-line;    /* Collapses whitespace, preserves line breaks */
white-space: break-spaces; /* Like pre-wrap but breaks at spaces */

/* Text overflow */
overflow: hidden;
text-overflow: ellipsis;  /* Shows ... when text overflows */
white-space: nowrap;

/* Multi-line ellipsis (webkit only) */
display: -webkit-box;
-webkit-line-clamp: 3;
-webkit-box-orient: vertical;
overflow: hidden;

/* Text shadow */
text-shadow: 2px 2px 4px rgba(0,0,0,0.3);
text-shadow: 1px 1px 0 #000, -1px -1px 0 #000; /* Outline effect */
text-shadow: 0 0 10px #3b82f6, 0 0 20px #3b82f6; /* Glow effect */

/* Text direction */
direction: rtl;           /* Right-to-left */
unicode-bidi: bidi-override;

/* Writing mode (vertical text) */
writing-mode: horizontal-tb;  /* Default */
writing-mode: vertical-rl;    /* Vertical, right-to-left */
writing-mode: vertical-lr;    /* Vertical, left-to-right;

/* Text orientation */
text-orientation: mixed;      /* Default for vertical */
text-orientation: upright;    /* Characters upright in vertical */

/* Font smoothing (webkit/macOS) */
-webkit-font-smoothing: antialiased;
-moz-osx-font-smoothing: grayscale;

/* Font display (for @font-face) */
font-display: swap;       /* Show fallback font immediately, swap when loaded */
font-display: block;      /* Hide text briefly, then show custom font */
font-display: fallback;   /* Brief swap period, then use fallback */
font-display: optional;   /* Use custom font only if already cached */
```

@font-face & Web Fonts Custom Fonts

```
/* Define custom font */
@font-face {
    font-family: 'MyFont';
    src: url('myfont.woff2') format('woff2'),
         url('myfont.woff') format('woff');
    font-weight: 400;
    font-style: normal;
    font-display: swap;  /* Prevents invisible text during load */
}

@font-face {
    font-family: 'MyFont';
    src: url('myfont-bold.woff2') format('woff2');
    font-weight: 700;
    font-style: normal;
}

/* Use the font */
body {
    font-family: 'MyFont', Arial, sans-serif;
}

/* Variable fonts (single file, multiple weights/styles) */
@font-face {
    font-family: 'Inter';
    src: url('Inter-Variable.woff2') format('woff2-variations');
    font-weight: 100 900;  /* Range of available weights */
    font-style: oblique 0deg 10deg;
}

/* Using variable font */
.thin { font-weight: 100; }
.regular { font-weight: 400; }
.bold { font-weight: 700; }
.black { font-weight: 900; }

/* Google Fonts import */
@import url('https://fonts.googleapis.com/css2?family=Inter:wght@400;600;700&display=swap');

/* Or link in HTML head */
<link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;600;700&display=swap" rel="stylesheet">
```

> **Font format priority:** WOFF2 (best compression) → WOFF (wide support) → TTF/OTF (fallback). Always include `font-display: swap` to prevent invisible text during font loading (FOIT).

14. CSS Flexbox

Flexbox Container Properties `display: flex`

```
.container {
    display: flex;           /* or inline-flex */

    /* Direction: main axis */
    flex-direction: row;              /* Default: left to right */
    flex-direction: row-reverse;      /* Right to left */
    flex-direction: column;           /* Top to bottom */
    flex-direction: column-reverse;   /* Bottom to top */

    /* Wrapping */
    flex-wrap: nowrap;        /* Default: single line */
    flex-wrap: wrap;          /* Multiple lines, top to bottom */
    flex-wrap: wrap-reverse;  /* Multiple lines, bottom to top */

    /* Shorthand: direction + wrap */
    flex-flow: row wrap;

    /* Main axis alignment */
    justify-content: flex-start;     /* Default */
    justify-content: flex-end;       /* Align to end */
    justify-content: center;         /* Center */
    justify-content: space-between;  /* Even space BETWEEN items */
    justify-content: space-around;   /* Even space AROUND items (including edges) */
    justify-content: space-evenly;   /* Truly even space (including edges) */
    justify-content: start;          /* Logical start (respects writing mode) */
    justify-content: end;            /* Logical end */
    justify-content: left;           /* Physical left */
    justify-content: right;          /* Physical right */

    /* Cross axis alignment (single line) */
    align-items: stretch;     /* Default: stretch to fill */
    align-items: flex-start;  /* Align to cross-start */
    align-items: flex-end;    /* Align to cross-end */
    align-items: center;      /* Center on cross axis */
    align-items: baseline;    /* Align text baselines */

    /* Cross axis alignment (multiple lines) */
    align-content: stretch;     /* Default */
    align-content: flex-start;
    align-content: flex-end;
    align-content: center;
    align-content: space-between;
    align-content: space-around;
    align-content: space-evenly;

    /* Gap between items */
    gap: 20px;                /* Row and column gap */
    gap: 10px 20px;         /* Row gap | Column gap */
    row-gap: 10px;
    column-gap: 20px;
}
```

Flexbox Item Properties flex items

```
.item {
    /* Order (default 0, higher = later) */
    order: 1;

    /* Flex grow: ability to grow (default 0) */
    flex-grow: 1;           /* Will grow to fill space */
    flex-grow: 0;           /* Will NOT grow (default) */

    /* Flex shrink: ability to shrink (default 1) */
    flex-shrink: 1;         /* Will shrink if needed (default) */
    flex-shrink: 0;         /* Will NOT shrink */

    /* Flex basis: starting size before grow/shrink */
    flex-basis: auto;       /* Based on content size */
    flex-basis: 200px;      /* Fixed starting size */
    flex-basis: 30%;        /* Percentage of container */
    flex-basis: 0;          /* Ignores content, sizes purely by flex-grow */

    /* Shorthand: grow shrink basis */
    flex: 1;                /* 1 1 0 (grow, shrink, basis: 0) */
    flex: 1 1 auto;         /* grow, shrink, basis: auto */
    flex: 0 0 200px;        /* Fixed 200px, no grow/shrink */
    flex: none;             /* 0 0 auto (fixed size) */
    flex: auto;             /* 1 1 auto (flexible) */

    /* Self alignment (overrides align-items for this item) */
    align-self: auto;       /* Inherit from container */
    align-self: flex-start;
    align-self: flex-end;
    align-self: center;
    align-self: baseline;
    align-self: stretch;
}
```

Common Flexbox Patterns: `flex: 1` makes items share space equally. `flex: 0 0 200px` creates fixed-width sidebar. `margin-left: auto` on a flex item pushes it to the right.

Flexbox Common Patterns Recipes

```
/* Center anything (horizontal + vertical) */
.center-container {
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
}

/* Equal height columns */
.columns {
    display: flex;
    gap: 20px;
}
.columns > * {
    flex: 1;
}

/* Sidebar + Main content */
.layout {
    display: flex;
    gap: 20px;
}
.sidebar {
    flex: 0 0 250px;    /* Fixed 250px sidebar */
}
.main {
    flex: 1;            /* Takes remaining space */
}

/* Header with logo left, nav right */
.header {
    display: flex;
    justify-content: space-between;
    align-items: center;
}

/* Push last item to right */
.nav {
    display: flex;
    gap: 20px;
}
.nav .push-right {
    margin-left: auto;
}

/* Wrap grid with flexbox */
.card-grid {
    display: flex;
    flex-wrap: wrap;
    gap: 20px;
}
.card-grid > * {
    flex: 1 1 300px;    /* Grow, shrink, basis 300px */
}

/* Sticky footer */
body {
    display: flex;
    flex-direction: column;
    min-height: 100vh;
}
main {
    flex: 1;            /* Pushes footer down */
}
```

15. CSS Grid

Grid Container Properties `display: grid`

```
.grid-container {
    display: grid;           /* or inline-grid */

    /* Define columns */
    grid-template-columns: 200px 200px 200px;     /* 3 fixed columns */
    grid-template-columns: 1fr 1fr 1fr;            /* 3 equal columns */
    grid-template-columns: repeat(3, 1fr);         /* Same as above */
    grid-template-columns: 2fr 1fr 1fr;            /* 2:1:1 ratio */
    grid-template-columns: 200px 1fr 200px;        /* Fixed + Flexible + Fixed */
    grid-template-columns: repeat(auto-fit, minmax(250px, 1fr)); /* Responsive! */
    grid-template-columns: repeat(auto-fill, minmax(250px, 1fr));

    /* Define rows */
    grid-template-rows: 100px auto 100px;          /* Fixed, auto, fixed */
    grid-template-rows: repeat(2, 1fr);            /* 2 equal rows */

    /* Named grid areas (visual layout) */
    grid-template-areas:
        "header header header"
        "sidebar main main"
        "footer footer footer";

    /* Gap between cells */
    gap: 20px;               /* Row and column gap */
    gap: 10px 20px;         /* Row | Column */
    row-gap: 10px;
    column-gap: 20px;

    /* Align items within cells (horizontal) */
    justify-items: stretch;   /* Default: fill cell width */
    justify-items: start;     /* Align to left */
    justify-items: end;       /* Align to right */
    justify-items: center;    /* Center horizontally */

    /* Align items within cells (vertical) */
    align-items: stretch;     /* Default: fill cell height */
    align-items: start;       /* Align to top */
    align-items: end;         /* Align to bottom */
    align-items: center;      /* Center vertically */

    /* Shorthand: align-items + justify-items */
    place-items: center;      /* Center both axes */
    place-items: start center; /* Vertical | Horizontal */

    /* Align entire grid (horizontal) */
    justify-content: start;   /* Default */
    justify-content: end;
    justify-content: center;
    justify-content: space-between;
    justify-content: space-around;
    justify-content: space-evenly;

    /* Align entire grid (vertical) */
    align-content: start;
    align-content: end;
    align-content: center;
    align-content: space-between;
    align-content: space-around;
    align-content: space-evenly;

    /* Shorthand: align-content + justify-content */
    place-content: center;

    /* Implicit grid (for items outside defined grid) */
    grid-auto-columns: 200px;
    grid-auto-rows: 100px;
    grid-auto-flow: row;      /* Default: fill row by row */
    grid-auto-flow: column;   /* Fill column by column */
    grid-auto-flow: row dense; /* Fill holes (dense packing) */
}
```

Grid Item Properties grid items

```
.grid-item {
    /* Position by line numbers */
    grid-column-start: 1;
    grid-column-end: 3;       /* Spans columns 1 to 3 (but not including 3) */
    grid-row-start: 1;
    grid-row-end: 2;

    /* Shorthand: start / end */
    grid-column: 1 / 3;       /* Same as above */
    grid-column: 1 / span 2;  /* Start at 1, span 2 columns */
    grid-column: span 2;      /* Span 2 columns from auto position */
    grid-column: 1 / -1;      /* From 1 to last column */
    grid-row: 1 / 2;

    /* Even shorter: row-start / column-start / row-end / column-end */
    grid-area: 1 / 1 / 2 / 3;

    /* Position by named area */
    grid-area: header;        /* Matches grid-template-areas */

    /* Self alignment (overrides container) */
    justify-self: center;     /* Horizontal alignment in cell */
    align-self: center;       /* Vertical alignment in cell */
    place-self: center;       /* Both axes */
    place-self: end center;   /* Vertical | Horizontal */

    /* Layering (z-index equivalent) */
    z-index: 2;               /* Higher = on top */
    order: 1;                 /* Like flexbox order */
}
```

```
/* Complete Grid Layout Example */
.layout {
    display: grid;
    grid-template-columns: 250px 1fr;
    grid-template-rows: 60px 1fr 40px;
    grid-template-areas:
        "header header"
        "sidebar main"
        "footer footer";
    gap: 20px;
    min-height: 100vh;
}

.header { grid-area: header; }
.sidebar { grid-area: sidebar; }
.main { grid-area: main; }
.footer { grid-area: footer; }

/* Responsive: stack on mobile */
@media (max-width: 768px) {
    .layout {
        grid-template-columns: 1fr;
        grid-template-rows: auto;
        grid-template-areas:
            "header"
            "main"
            "sidebar"
            "footer";
    }
}
```

Grid vs Flexbox - When to Use What Decision Guide

Use Flexbox when...	Use Grid when...
Single dimension (row OR column)	Two dimensions (rows AND columns)
Content determines layout	Layout determines content placement
Aligning items in a line	Creating complex page layouts
Navigation bars, card footers	Page layouts, photo galleries
Centering single elements	Overlapping content areas
Equal height columns	Asymmetric layouts

> **They work great together!** Use Grid for the overall page layout, then Flexbox inside grid cells for component-level alignment.

16. CSS Positioning

Position Property Complete Guide position

```
/* static: Default, normal document flow */
.static {
    position: static;
    /* top, right, bottom, left, z-index have NO effect */
}

/* relative: Positioned relative to normal position */
.relative {
    position: relative;
    top: 10px;        /* Moves DOWN 10px from normal position */
    left: 20px;       /* Moves RIGHT 20px from normal position */
    /* Element still takes up original space in layout */
}

/* absolute: Positioned relative to nearest positioned ancestor */
/* If no positioned ancestor, uses initial containing block (viewport) */
.parent {
    position: relative;  /* Creates positioning context */
}
.child {
    position: absolute;
    top: 0;           /* 0px from top of parent */
    right: 0;         /* 0px from right of parent */
    bottom: 20px;     /* 20px from bottom of parent */
    left: 20px;       /* 20px from left of parent */
    /* Removed from document flow - other elements act like it's not there */
}

/* fixed: Positioned relative to viewport */
/* Stays in place when scrolling */
.fixed {
    position: fixed;
    top: 20px;
    right: 20px;
    /* Removed from document flow */
}

/* sticky: Hybrid of relative and fixed */
/* Acts like relative until scroll threshold, then sticks */
.sticky {
    position: sticky;
    top: 0;           /* Sticks when element reaches top of viewport */
    /* Still takes up space in layout */
}

/* Common sticky patterns */
.sticky-header {
    position: sticky;
    top: 0;
    z-index: 100;
}

.sticky-sidebar {
    position: sticky;
    top: 80px;        /* 80px below top (after header) */
    align-self: start; /* Required in flex/grid containers */
}

/* Inset shorthand (modern) */
.inset {
    position: absolute;
    inset: 0;              /* top:0 right:0 bottom:0 left:0 (fill parent) */
    inset: 10px;           /* All sides 10px */
    inset: 10px 20px;      /* Vertical | Horizontal */
    inset: 10px 20px 30px 40px; /* Top Right Bottom Left */
}

/* Centering with absolute positioning */
.center-absolute {
    position: absolute;
    top: 50%;
    left: 50%;
    transform: translate(-50%, -50%);  /* Offset by half its own size */
}

/* Another centering method */
```

```
.center-inset {
    position: absolute;
    inset: 0;
    margin: auto;
    width: fit-content;
    height: fit-content;
}
```

Position	Reference	In flow?	Scrolls?
static	Normal flow	Yes	Yes
relative	Normal position	Yes	Yes
absolute	Positioned ancestor	No	Yes
fixed	Viewport	No	No
sticky	Scroll position	Yes	Yes (until sticky)

Z-Index & Stacking Context z-index

```
/* z-index only works on positioned elements */
/* (position: relative/absolute/fixed/sticky) or flex/grid items */

.box1 { z-index: 1; }
.box2 { z-index: 2; }  /* Appears on top of box1 */
.box3 { z-index: 999; } /* Highest in this context */

/* Creating new stacking contexts */
.new-context {
    /* Any of these create a new stacking context: */
    position: relative; z-index: 0;
    position: relative; z-index: auto; /* Does NOT create context */
    opacity: 0.99;
    transform: translate(0);
    filter: blur(0);
    perspective: 1px;
    isolation: isolate;       /* Explicitly creates context */
    will-change: transform;
    contain: layout;
}

/* Stacking order within a context (lowest to highest):
   1. Root element background/border
   2. Positioned elements with z-index < 0
   3. Non-positioned block elements
   4. Non-positioned floats
   5. Non-positioned inline elements
   6. Positioned elements with z-index: auto / 0
   7. Positioned elements with z-index > 0
*/

/* Best practice: Use isolation */
.component {
    isolation: isolate;  /* Prevents z-index leaking */
    position: relative;
    z-index: 1;
}
```

> A common z-index bug: An element with `z-index: 999` can still appear BELOW an element with `z-index: 1` if they're in different stacking contexts. The z-index only competes within the same stacking context.

17. Responsive Design

Media Queries @media

```
/* Mobile-first approach (recommended) */
/* Base styles for mobile */
.container {
    width: 100%;
    padding: 1rem;
}

/* Tablet (768px and up) */
@media (min-width: 768px) {
    .container {
        width: 750px;
        margin: 0 auto;
    }
}

/* Desktop (1024px and up) */
@media (min-width: 1024px) {
    .container {
        width: 960px;
    }
}

/* Large desktop (1280px and up) */
@media (min-width: 1280px) {
    .container {
        width: 1200px;
    }
}

/* Range syntax (modern browsers) */
@media (768px <= width < 1024px) {
    .sidebar { display: none; }
}

/* Common breakpoints */
/* Mobile: < 640px */
/* Tablet: 640px - 1024px */
/* Desktop: > 1024px */

/* Orientation */
@media (orientation: landscape) {
    .hero { height: 60vh; }
}
@media (orientation: portrait) {
    .hero { height: 40vh; }
}

/* Aspect ratio */
@media (min-aspect-ratio: 16/9) {
    .video { aspect-ratio: 16/9; }
}

/* Device features */
@media (hover: hover) {
    /* Device has a pointing device (mouse) */
    .btn:hover { background: #1d4ed8; }
}
@media (hover: none) {
    /* Touch device - no hover */
    .btn:active { background: #1d4ed8; }
}

/* Pointer accuracy */
@media (pointer: coarse) {
    /* Touch screen - larger tap targets */
    .btn { min-height: 44px; min-width: 44px; }
}
@media (pointer: fine) {
    /* Mouse/stylus - precise targeting */
    .btn { min-height: 32px; }
}

/* Dark mode */
@media (prefers-color-scheme: dark) {
    body {
        background: #0f172a;
```

```
        color: #e2e8f0;
    }
}

/* Reduced motion (accessibility) */
@media (prefers-reduced-motion: reduce) {
    *, *::before, *::after {
        animation-duration: 0.01ms !important;
        animation-iteration-count: 1 !important;
        transition-duration: 0.01ms !important;
    }
}

/* High contrast mode */
@media (prefers-contrast: high) {
    .btn {
        border: 2px solid currentColor;
    }
}

/* Print styles */
@media print {
    .no-print { display: none !important; }
    body { font-size: 12pt; color: black; }
    a { text-decoration: underline; color: black; }
    a[href]::after { content: " (" attr(href) ")"; font-size: 10pt; }
    .page-break { page-break-before: always; }
}

/* Container queries (modern - element-based responsive) */
/* Define container */
.card-container {
    container-type: inline-size;
    container-name: card;
}

/* Query based on container width, not viewport */
@container card (min-width: 400px) {
    .card { display: flex; }
    .card-image { width: 40%; }
}

@container card (min-width: 600px) {
    .card { flex-direction: row; }
    .card-title { font-size: 1.5rem; }
}
```

> **Mobile-first vs Desktop-first:** Mobile-first uses `min-width` (start small, add complexity). Desktop-first uses `max-width` (start large, remove complexity). Mobile-first is generally preferred because it's easier to add complexity than to remove it.

Responsive Images & Containers Responsive

```
/* Responsive container */
.container {
    width: min(100% - 2rem, 1200px);
    margin-inline: auto;
}

/* Fluid typography */
h1 {
    font-size: clamp(2rem, 5vw + 1rem, 4rem);
}

/* Responsive padding */
.section {
    padding: clamp(1rem, 5vw, 4rem);
}

/* Aspect ratio boxes */
.video-container {
    aspect-ratio: 16 / 9;
    width: 100%;
}

/* Responsive grid */
.grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(min(100%, 300px), 1fr));
    gap: clamp(1rem, 3vw, 2rem);
}

/* Container query example */
.product-card {
    container-type: inline-size;
}

@container (min-width: 350px) {
    .product-card {
        display: grid;
        grid-template-columns: 120px 1fr;
        gap: 1rem;
    }
}
```

18. CSS Effects

Transitions `transition`

```
/* Shorthand: property duration timing-function delay */
.btn {
    background: #3b82f6;
    transition: all 0.3s ease;
    /* property: all (any changing property) */
    /* duration: 0.3s */
    /* timing-function: ease */
    /* delay: 0s */
}

.btn:hover {
    background: #1d4ed8;
    transform: translateY(-2px);
    box-shadow: 0 4px 12px rgba(0,0,0,0.15);
}

/* Individual properties */
.animated {
    transition-property: background, transform, opacity;
    transition-duration: 0.3s, 0.5s, 0.2s;
    transition-timing-function: ease-in-out, cubic-bezier(0.4, 0, 0.2, 1), linear;
    transition-delay: 0s, 0.1s, 0s;
}

/* Timing functions */
/* ease: Slow start, fast middle, slow end (default) */
/* linear: Constant speed */
/* ease-in: Slow start */
/* ease-out: Slow end */
/* ease-in-out: Slow start and end */
/* cubic-bezier(x1, y1, x2, y2): Custom curve */
/* steps(n): Discrete steps */

/* Common cubic-bezier values */
.cubic-smooth { transition-timing-function: cubic-bezier(0.4, 0, 0.2, 1); }
.cubic-decelerate { transition-timing-function: cubic-bezier(0, 0, 0.2, 1); }
.cubic-accelerate { transition-timing-function: cubic-bezier(0.4, 0, 1, 1); }
.cubic-sharp { transition-timing-function: cubic-bezier(0.4, 0, 0.6, 1); }

/* Multiple transitions */
.complex {
    transition:
        background 0.3s ease,
        transform 0.5s cubic-bezier(0.4, 0, 0.2, 1) 0.1s,
        opacity 0.2s linear;
}

/* Transition specific properties */
.specific {
    transition-property: opacity, visibility;
    transition-duration: 0.3s;
}

/* Will-change (performance hint) */
.animated-element {
    will-change: transform, opacity;
    /* Remove after animation to free resources */
}

/* Hardware acceleration */
.gpu-accelerated {
    transform: translateZ(0);  /* Forces GPU compositing */
    backface-visibility: hidden;
}
```

Animations & @keyframes `@keyframes`

```
/* Define keyframes */
@keyframes fadeIn {
    from { opacity: 0; }
    to { opacity: 1; }
```

```
}

@keyframes slideIn {
    0% {
        transform: translateX(-100%);
        opacity: 0;
    }
    100% {
        transform: translateX(0);
        opacity: 1;
    }
}

@keyframes bounce {
    0%, 100% { transform: translateY(0); }
    50% { transform: translateY(-20px); }
}

@keyframes spin {
    from { transform: rotate(0deg); }
    to { transform: rotate(360deg); }
}

@keyframes pulse {
    0%, 100% { transform: scale(1); opacity: 1; }
    50% { transform: scale(1.05); opacity: 0.8; }
}

@keyframes shake {
    0%, 100% { transform: translateX(0); }
    25% { transform: translateX(-10px); }
    75% { transform: translateX(10px); }
}

@keyframes gradientShift {
    0% { background-position: 0% 50%; }
    50% { background-position: 100% 50%; }
    100% { background-position: 0% 50%; }
}

/* Apply animation */
.element {
    animation: fadeIn 0.5s ease-out;

    /* Individual properties */
    animation-name: fadeIn;
    animation-duration: 0.5s;
    animation-timing-function: ease-out;
    animation-delay: 0s;
    animation-iteration-count: 1;       /* or infinite */
    animation-direction: normal;        /* normal | reverse | alternate | alternate-reverse */
    animation-fill-mode: forwards;      /* none | forwards | backwards | both */
    animation-play-state: running;      /* running | paused */
}

/* Common animation patterns */
.infinite-spin {
    animation: spin 1s linear infinite;
}

.pulse {
    animation: pulse 2s ease-in-out infinite;
}

.slide-in {
    animation: slideIn 0.5s cubic-bezier(0.4, 0, 0.2, 1) forwards;
}

/* Staggered animations */
.stagger-1 { animation-delay: 0.1s; }
.stagger-2 { animation-delay: 0.2s; }
.stagger-3 { animation-delay: 0.3s; }

/* Paused on hover */
.animated:hover {
    animation-play-state: paused;
}

/* Fill modes explained */
/* none: Default, returns to original state */
/* forwards: Keeps final keyframe state */
/* backwards: Applies first keyframe before delay */
/* both: Applies both forwards and backwards */

/* Shorthand */
```

```
.animation-shorthand {
    animation: slideIn 0.5s ease-out 0.2s 1 forwards;
    /* name duration timing-function delay iteration-count fill-mode */
}
```

Transforms transform

```
/* 2D Transforms */
.translate {
    transform: translate(50px, 100px);    /* X, Y */
    transform: translateX(20px);
    transform: translateY(-10px);
    transform: translate(50%, 50%);       /* Percentage of element size */
}

.scale {
    transform: scale(1.5);                /* 150% size */
    transform: scale(0.5);                /* 50% size */
    transform: scale(1, 2);               /* scaleX(1) scaleY(2) */
    transform: scaleX(2);
    transform: scaleY(0.5);
}

.rotate {
    transform: rotate(45deg);
    transform: rotate(-15deg);
    transform: rotate(0.5turn);           /* 180 degrees */
    transform: rotate(1rad);              /* Radians */
}

.skew {
    transform: skew(10deg, 5deg);         /* X, Y */
    transform: skewX(10deg);
    transform: skewY(5deg);
}

/* Multiple transforms (order matters!) */
.combined {
    transform: translateX(20px) rotate(15deg) scale(1.1);
    /* Applied right to left: scale → rotate → translate */
}

/* Transform origin */
.origin-center {
    transform-origin: center;             /* Default */
}
.origin-corner {
    transform-origin: top left;
    transform-origin: bottom right;
    transform-origin: 20% 80%;           /* Percentages */
    transform-origin: 50px 100px;        /* Pixel values */
}

/* 3D Transforms */
.perspective-container {
    perspective: 1000px;                  /* Depth of 3D space */
    perspective-origin: center;
}

.card-3d {
    transform-style: preserve-3d;        /* Children exist in 3D space */
    transform: rotateY(45deg);
}

.translate-3d {
    transform: translate3d(0, 0, 100px);  /* X, Y, Z */
}

.rotate-3d {
    transform: rotateX(45deg);
    transform: rotateY(45deg);
    transform: rotateZ(45deg);
    transform: rotate3d(1, 1, 1, 45deg); /* X, Y, Z axis + angle */
}

.scale-3d {
    transform: scale3d(1, 1, 2);
}

/* Backface visibility */
```

```
.flippable {
    backface-visibility: hidden;          /* Hidden when rotated away */
}

/* Hardware acceleration trick */
.gpu {
    transform: translateZ(0);
    will-change: transform;
}
```

Box Shadow & Filters Visual Effects

```
/* Box Shadow */
/* Syntax: offset-x offset-y blur-radius spread-radius color */
.box {
    box-shadow: 2px 2px 5px rgba(0,0,0,0.3);

    /* Multiple shadows (layered) */
    box-shadow:
        0 1px 2px rgba(0,0,0,0.1),
        0 4px 8px rgba(0,0,0,0.1),
        0 16px 32px rgba(0,0,0,0.1);

    /* Inset shadow (inside element) */
    box-shadow: inset 0 2px 4px rgba(0,0,0,0.1);

    /* Colored shadow */
    box-shadow: 0 10px 30px rgba(59, 130, 246, 0.3);

    /* Neumorphism (soft UI) */
    box-shadow:
        8px 8px 16px #d1d5db,
        -8px -8px 16px #ffffff;

    /* Inner glow */
    box-shadow: inset 0 0 20px rgba(59, 130, 246, 0.5);
}

/* Drop shadow (follows transparent edges of PNG/SVG) */
.image {
    filter: drop-shadow(0 4px 8px rgba(0,0,0,0.3));
    /* Unlike box-shadow, follows alpha channel! */
}

/* CSS Filters */
.filter {
    filter: blur(5px);
    filter: brightness(150%);
    filter: contrast(200%);
    filter: grayscale(100%);
    filter: sepia(100%);
    filter: hue-rotate(90deg);
    filter: invert(100%);
    filter: saturate(200%);
    filter: opacity(50%);

    /* Multiple filters */
    filter: grayscale(50%) blur(2px) brightness(120%);

    /* Backdrop filter (blurs content BEHIND element) */
    backdrop-filter: blur(10px);
    backdrop-filter: saturate(180%) blur(20px);
    background: rgba(255,255,255,0.7); /* Semi-transparent background needed */
}

/* Clip path */
.clipped {
    clip-path: circle(50%);                    /* Circle */
    clip-path: circle(50% at 30% 70%);        /* Circle with offset center */
    clip-path: ellipse(50% 30%);              /* Ellipse */
    clip-path: polygon(0 0, 100% 0, 50% 100%); /* Triangle */
    clip-path: inset(10px 20px 30px 40px);    /* Inset rectangle */
    clip-path: inset(10px round 8px);         /* With border radius */
}

/* Mask (uses image as mask) */
.masked {
    mask-image: url('mask.png');
    mask-size: cover;
    -webkit-mask-image: url('mask.png');      /* Webkit prefix needed */
    -webkit-mask-size: cover;
}
```

19. Advanced CSS

CSS Custom Properties (Variables) --var

```
/* Define variables in :root (global scope) */
:root {
    --primary-color: #3b82f6;
    --secondary-color: #8b5cf6;
    --text-color: #1e293b;
    --bg-color: #ffffff;
    --border-radius: 8px;
    --spacing-unit: 1rem;
    --font-stack: 'Segoe UI', system-ui, sans-serif;
    --shadow-sm: 0 1px 2px rgba(0,0,0,0.1);
    --shadow-md: 0 4px 6px rgba(0,0,0,0.1);
    --shadow-lg: 0 10px 25px rgba(0,0,0,0.1);
}

/* Scoped variables */
.card {
    --card-padding: 1.5rem;
    --card-bg: #ffffff;
    padding: var(--card-padding);
    background: var(--card-bg);
}

/* Use variables */
.btn-primary {
    background: var(--primary-color);
    color: white;
    border-radius: var(--border-radius);
    font-family: var(--font-stack);
    box-shadow: var(--shadow-md);
}

/* Fallback values */
.unknown {
    color: var(--undefined-var, #333);  /* Uses #333 if variable doesn't exist */
}

/* Dynamic theming with data attributes */
[data-theme="dark"] {
    --text-color: #e2e8f0;
    --bg-color: #0f172a;
    --card-bg: #1e293b;
}

/* JavaScript can modify variables */
/* document.documentElement.style.setProperty('--primary-color', '#ff0000'); */

/* Calc with variables */
.responsive {
    width: calc(100% - var(--spacing-unit) * 2);
    padding: calc(var(--spacing-unit) / 2);
}

/* Variable inheritance */
.parent {
    --local-color: red;
}
.child {
    color: var(--local-color);  /* Inherits from parent */
}

/* @property (registered custom properties) - for animations */
@property --progress {
    syntax: '<percentage>';
    inherits: false;
    initial-value: 0%;
}

.animated-progress {
    --progress: 0%;
    background: linear-gradient(90deg, #3b82f6 var(--progress), transparent var(--progress));
    animation: progress 2s ease forwards;
}

@keyframes progress {
    to { --progress: 100%; }
}
```

CSS Variables are scoped to the element they're defined on and inherited by children. They're incredibly powerful for theming, responsive design, and maintaining design systems. Unlike Sass/Less variables, they can be changed at runtime with JavaScript.

calc(), min(), max(), clamp() Math Functions

```
/* calc() - Calculate values */
.sidebar {
    width: calc(100% - 300px);           /* Subtract fixed width */
    width: calc(50% + 20px);             /* Mix units */
    width: calc(100vw - 2rem);          /* Viewport minus padding */
    width: calc(var(--base) * 2);       /* With variables */
    width: calc((100% - 40px) / 3);     /* Complex math */
    margin: calc(1rem * 2) calc(1rem / 2);
}

/* min() - Smallest of values */
.container {
    width: min(100%, 1200px);           /* Never wider than 1200px, but 100% on small screens */
    font-size: min(5vw, 3rem);         /* Caps font size */
}

/* max() - Largest of values */
.element {
    width: max(300px, 50%);            /* At least 300px, or 50% if larger */
    font-size: max(1rem, 2vw);        /* At least 1rem */
}

/* clamp() - Min, Preferred, Max */
.fluid {
    width: clamp(300px, 50%, 800px);   /* Min: 300px, Preferred: 50%, Max: 800px */
    font-size: clamp(1rem, 2.5vw, 2rem); /* Responsive typography */
    padding: clamp(1rem, 5vw, 3rem);    /* Responsive spacing */
}

/* Practical examples */
.responsive-card {
    /* Card that scales between 280px and 400px, preferring 30% of container */
    width: clamp(280px, 30%, 400px);
}

.hero-title {
    /* Fluid typography that never gets too small or too large */
    font-size: clamp(2rem, 5vw + 1rem, 4rem);
}

.sticky-footer {
    /* Footer always at least viewport height minus header */
    min-height: calc(100vh - 60px);
}
```

CSS Functions Reference Functions

Function	Description	Example
calc()	Mathematical calculations	calc(100% - 20px)
min()	Smallest value	min(100%, 800px)
max()	Largest value	max(200px, 50%)
clamp()	Constrained value	clamp(1rem, 2vw, 2rem)
var()	Custom property value	var(--color)
env()	Environment variables	env(safe-area-inset-top)
rgb() / rgba()	RGB color	rgb(255, 0, 0)
hsl() / hsla()	HSL color	hsl(0, 100%, 50%)
hwb()	HWB color	hwb(0 0% 0%)
lab()	LAB color	lab(50% 50 50)
lch()	LCH color	lch(50% 50 0)
oklch()	OKLCH color	oklch(60% 0.2 250)
color-mix()	Mix two colors	color-mix(in srgb, red, blue 50%)
contrast-color()	Auto contrast color	contrast-color(black)
url()	Reference external resource	url('image.jpg')
attr()	HTML attribute value	attr(data-label)
counter()	CSS counter value	counter(section)
counters()	Nested counters	counters(section, ".")
linear-gradient()	Linear gradient	linear-gradient(to right, red, blue)
radial-gradient()	Radial gradient	radial-gradient(circle, red, blue)
conic-gradient()	Conic gradient	conic-gradient(red, blue)
repeating-linear-gradient()	Repeating linear	repeating-linear-gradient(45deg, red, blue 10px)
repeating-radial-gradient()	Repeating radial	repeating-radial-gradient(red, blue 10px)

CSS Counters counter-reset

```
/* Automatic section numbering */
body {
    counter-reset: section;
}

h2 {
    counter-reset: subsection;
}

h2::before {
    counter-increment: section;
    content: "Section " counter(section) ": ";
}

h3::before {
    counter-increment: subsection;
    content: counter(section) "." counter(subsection) " ";
}

/* Nested list numbering */
ol {
    counter-reset: item;
    list-style: none;
}

li::before {
    counter-increment: item;
    content: counters(item, ".") " ";
}

/* Page numbering for print */
@page {
    @bottom-center {
        content: counter(page) " of " counter(pages);
    }
}
```

CSS @rules Reference @rules

@rule	Purpose
@charset	Character encoding
@import	Import external stylesheet
@media	Responsive conditions
@font-face	Custom font definition
@keyframes	Animation definition
@supports	Feature detection
@layer	Cascade layers
@container	Container queries
@property	Registered custom property
@scope	Scoped styles
@page	Print page styling
@counter-style	Custom list bullets
@namespace	XML namespace
@document (deprecated)	Document-specific styles

```
/* @supports - Feature detection */
```

```
@supports (display: grid) {
    .layout { display: grid; }
}

@supports not (display: grid) {
    .layout { display: flex; }
}

@supports (aspect-ratio: 1/1) {
    .square { aspect-ratio: 1/1; }
}

@supports (backdrop-filter: blur(10px)) {
    .glass { backdrop-filter: blur(10px); }
}

/* @layer - Cascade layers (controls specificity) */
@layer reset, base, components, utilities;

@layer reset {
    * { margin: 0; padding: 0; }
}

@layer base {
    body { font-family: sans-serif; line-height: 1.5; }
}

@layer components {
    .btn { padding: 0.5rem 1rem; border-radius: 4px; }
}

@layer utilities {
    .text-center { text-align: center; }
}

/* @scope - Scoped styles (modern) */
@scope (.card) {
    :scope { background: white; }
    h2 { font-size: 1.25rem; }
    p { color: #666; }
}

/* @counter-style - Custom list markers */
@counter-style thumbs {
    system: cyclic;
    symbols: "F44D" "F44E";
    suffix: " ";
}

ul.reactions { list-style: thumbs; }
```

CSS Specificity & Cascade Specificity

```
/* Specificity: (ID selectors, Class selectors, Type selectors) */

/* 0,0,1 - Type selector */
p { color: black; }

/* 0,1,0 - Class selector */
.intro { color: blue; }

/* 0,1,1 - Class + Type */
p.intro { color: green; }

/* 1,0,0 - ID selector */
#header { color: red; }

/* 1,0,1 - ID + Type */
nav#header { color: purple; }

/* Inline style: 1,0,0,0 */
/* <p style="color: pink"> - Wins over everything */

/* !important: Overrides everything (use sparingly!) */
.text { color: orange !important; }

/* :where() has ZERO specificity */
:where(.btn) { color: red; }  /* 0,0,0 - easily overridden */

/* :is() takes highest specificity of its arguments */
:is(h1, h2, .title) { color: blue; }  /* 0,1,1 (h2 contributes most) */

/* :not() adds specificity of its argument */
:not(.special) { color: gray; }  /* 0,1,0

/* :has() adds specificity */
article:has(img) { padding: 0; }  /* 0,0,2

/* Cascade layers override specificity */
@layer base { #id { color: red; } }     /* Layer: base */
@layer utilities { .class { color: blue; } }  /* Layer: utilities - WINS! */
/* Even though #id has higher specificity, utilities layer wins */
```

Selector	Specificity	Score
*	0,0,0	0
li	0,0,1	1
li::before	0,0,2	2
.class	0,1,0	10
li.class	0,1,1	11
#id	1,0,0	100
#id.class	1,1,0	110
style attribute	1,0,0,0	1000
!important	∞	∞

Avoid !important. It breaks the natural cascade and makes debugging difficult. If you find yourself using it, your specificity is likely too high elsewhere. Use cascade layers (@layer) or increase selector specificity instead.

CSS Reset & Normalize Patterns Best Practices

```css
/* Modern CSS Reset (recommended) */
*, *::before, *::after {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}

/* Remove default list styles */
ul, ol {
    list-style: none;
}

/* Remove default link styles */
a {
    text-decoration: none;
    color: inherit;
}

/* Make images responsive by default */
img, picture, video, canvas, svg {
    display: block;
    max-width: 100%;
}

/* Inherit fonts for form elements */
input, button, textarea, select {
    font: inherit;
}

/* Remove default button styles */
button {
    border: none;
    background: none;
    cursor: pointer;
}

/* Smooth scrolling */
html {
    scroll-behavior: smooth;
}

/* Respect reduced motion preference */
@media (prefers-reduced-motion: reduce) {
    html {
        scroll-behavior: auto;
    }
    *, *::before, *::after {
        animation-duration: 0.01ms !important;
        animation-iteration-count: 1 !important;
        transition-duration: 0.01ms !important;
    }
}

/* Remove tap highlight on mobile */
* {
    -webkit-tap-highlight-color: transparent;
}

/* Focus visible (modern focus styles) */
:focus-visible {
    outline: 2px solid #3b82f6;
    outline-offset: 2px;
}

/* Remove focus outline for mouse users (keep for keyboard) */
:focus:not(:focus-visible) {
    outline: none;
}
```

20. Quick Reference Tables

HTML5 Elements Quick List All Elements

Category	Elements
Document Metadata	<html>, <head>, <title>, <base>, <link>, <meta>, <style>
Sectioning Root	<body>
Content Sectioning	<article>, <aside>, <footer>, <header>, <h1>-<h6>, <main>, <nav>, <section>
Text Content	<blockquote>, <dd>, , <dl>, <dt>, <figcaption>, <figure>, <hr>, , , <p>, <pre>,
Inline Text	<a>, <abbr>, <b>, <bdi>, <bdo>, , <cite>, <code>, <data>, <dfn>, , <i>, <kbd>, <mark>, <q>, <rp>, <rt>, <ruby>, <s>, <samp>, <small>, , , <sub>, <sup>, <time>, <u>, <var>, <wbr>
Image & Media	<area>, <audio>, <img>, <map>, <track>, <video>
Embedded Content	<embed>, <iframe>, <object>, <param>, <picture>, <source>
Scripting	<canvas>, <noscript>, <script>
Demarcating Edits	<del>, <ins>
Table Content	<caption>, <col>, <colgroup>, <table>, <tbody>, <td>, <tfoot>, <th>, <thead>, <tr>
Forms	<button>, <datalist>, <fieldset>, <form>, <input>, <label>, <legend>, <meter>, <optgroup>, <option>, <output>, <progress>, <select>, <textarea>
Interactive	<details>, <dialog>, <menu>, <summary>
Web Components	<slot>, <template>

CSS Properties Quick List (Most Common) Properties

Category	Properties
Box Model	width, height, padding, margin, border, box-sizing, overflow, aspect-ratio
Display	display, visibility, opacity, position, z-index, top, right, bottom, left
Flexbox	flex, flex-direction, flex-wrap, justify-content, align-items, align-content, gap, order, align-self
Grid	grid, grid-template-columns, grid-template-rows, grid-template-areas, grid-gap, justify-items, align-items, place-items, grid-column, grid-row, grid-area
Colors	color, background, background-color, background-image, background-size, background-position, background-repeat, opacity
Typography	font-family, font-size, font-weight, font-style, line-height, text-align, text-decoration, text-transform, letter-spacing, word-spacing, white-space, text-overflow
Transforms	transform, transform-origin, transform-style, perspective, backface-visibility
Transitions	transition, transition-property, transition-duration, transition-timing-function, transition-delay
Animations	animation, animation-name, animation-duration, animation-timing-function, animation-delay, animation-iteration-count, animation-direction, animation-fill-mode, animation-play-state
Effects	box-shadow, text-shadow, filter, backdrop-filter, clip-path, mask
Responsive	@media, container-type, @container, clamp(), min(), max()
Other	cursor, pointer-events, user-select, resize, scroll-behavior, object-fit, object-position

CSS Units Reference Units

Unit	Type	Description
px	Absolute	Pixels (device-independent)
cm, mm, in	Absolute	Centimeters, millimeters, inches
pt, pc	Absolute	Points (1/72in), picas (12pt)
em	Relative	Relative to parent font-size
rem	Relative	Relative to root (html) font-size
%	Relative	Percentage of parent
vw	Viewport	1% of viewport width
vh	Viewport	1% of viewport height
vmin	Viewport	Smaller of vw/vh
vmax	Viewport	Larger of vw/vh
svw, svh	Viewport	Small viewport (mobile, excludes UI)
lvw, lvh	Viewport	Large viewport (includes UI)
dvw, dvh	Viewport	Dynamic viewport (changes with UI)
ex	Font	Height of lowercase 'x'
ch	Font	Width of '0' character
cap	Font	Height of capital letters
lh	Font	Line height of element
rlh	Font	Line height of root element